

Machine-Learning-Enhanced Analysis of the Ising Model on Hyperbolic Lattices

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Overview

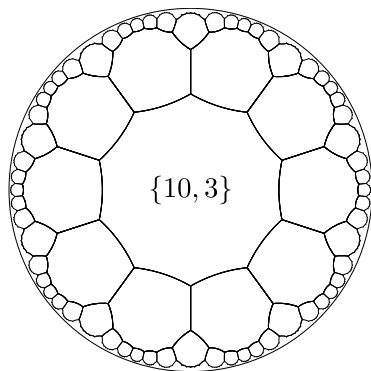
- 1 Hyperbolic Lattice Basics
- 2 Methodology - Machine Learning
- 3 Results
- 4 Conclusions

Outline

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Hyperbolic Lattices

- Tiling of negatively-curved surface
- Poincare Disk



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Multi-layer Perceptron (MLP)



- Simplest model
- Input: Spin configuration
- Output: $T < T_{\text{crit}}$ or $T > T_{\text{crit}}$

Dense MLP

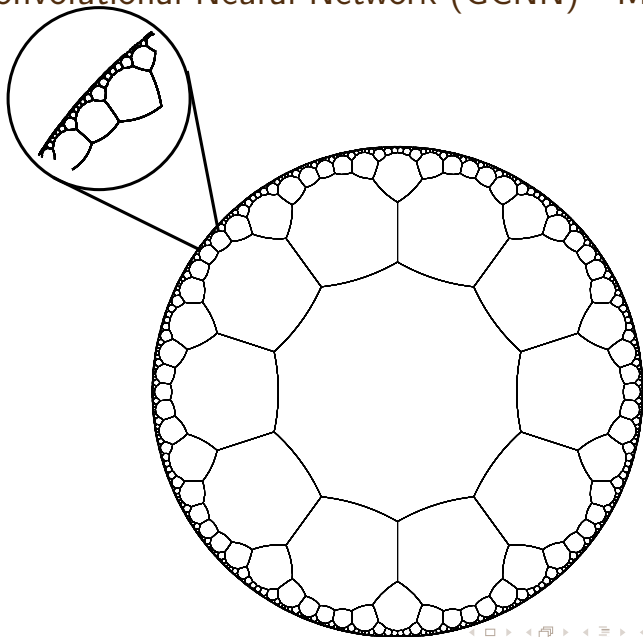
200 Neurons

Single hidden layer (ReLU)

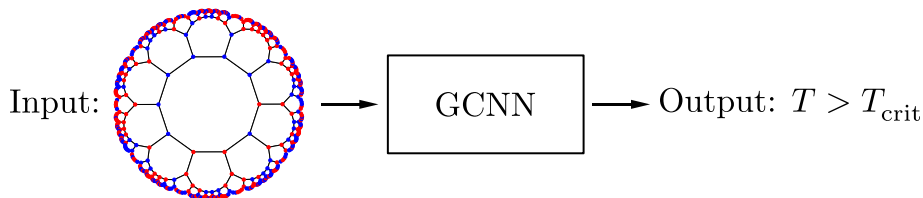
Softmax output

NAdam Optimizer

Graph Convolutional Neural Network (GCNN) - Motivation



Graph Convolutional Neural Network (GCNN)



- PyTorch support
- Input: hyperbolic lattice graphs
- Node features: spin state
- Output: Graph classification

GCNN

3 GCN Layers
200 Feature Maps Each
Softmax output
NAdam Optimizer

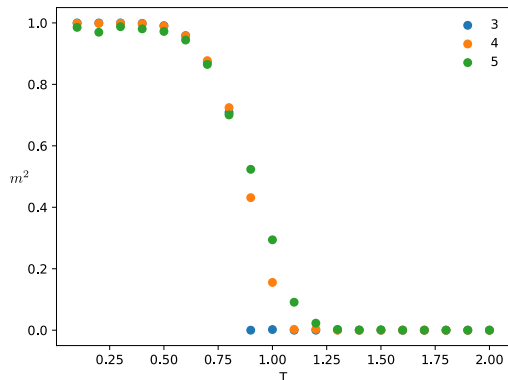
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Finding T_{crit}

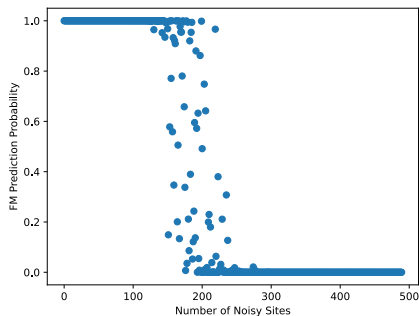
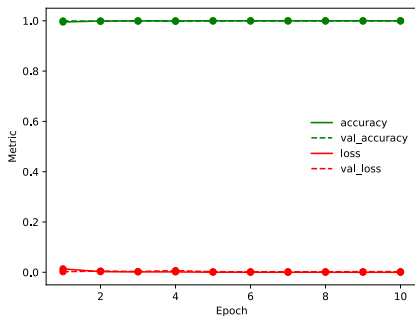
$$m = \frac{1}{N} \sum_i^N \sigma_i$$

- Magnetization (m)
- FM $\rightarrow m^2 = 1$
- PM $\rightarrow 0$
- $T_{\text{crit}} \approx 0.9$

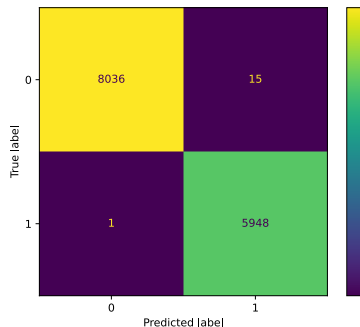


Rotationally-Variant Initialization

Ferromagnet $\longleftrightarrow$ All up spin



Rotationally-Variant Initialization



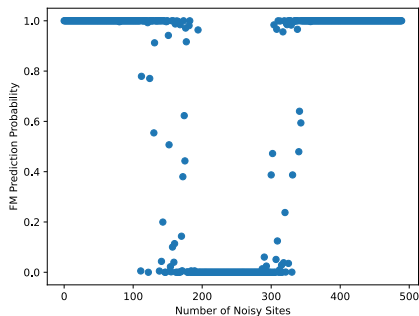
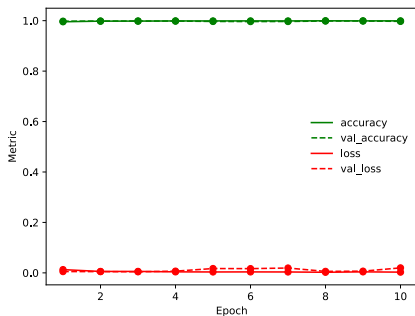
Test Data Accuracy: 0.999

Test Data Loss: 0.004

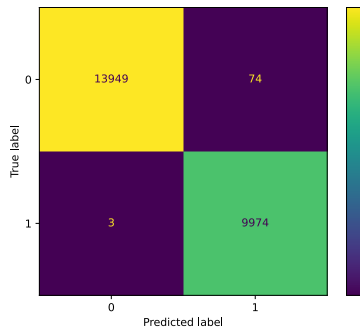
*Crucial detail regarding autocorrelations

Rotationally-Invariant Initialization

Ferromagnet $\iff$ All up spin or All down spin



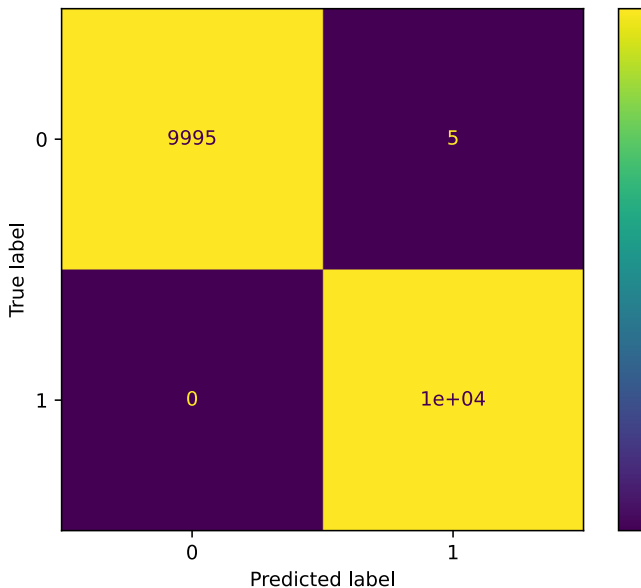
Rotationally-Invariant Initialization



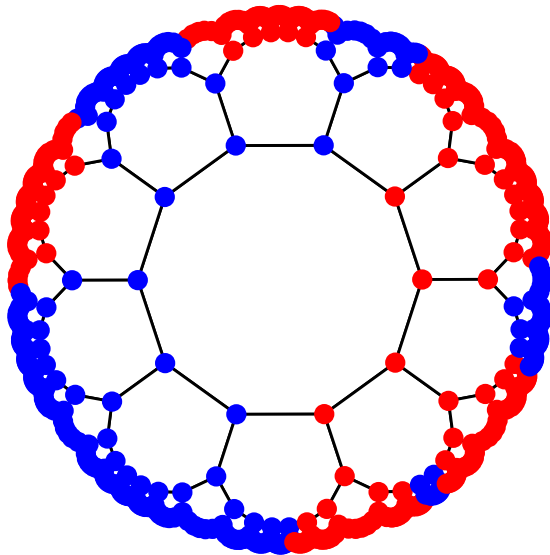
Test Data Accuracy: 0.997

Test Data Loss: 0.027

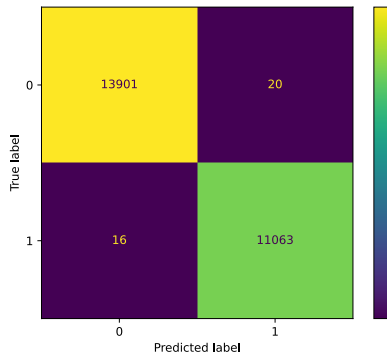
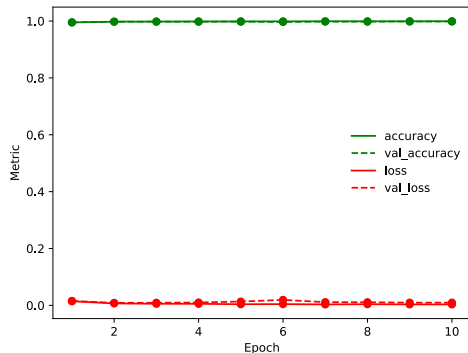
Rotationally-Invariant Initialization - Unseen MC Run



Domain Walls - Random Initialization



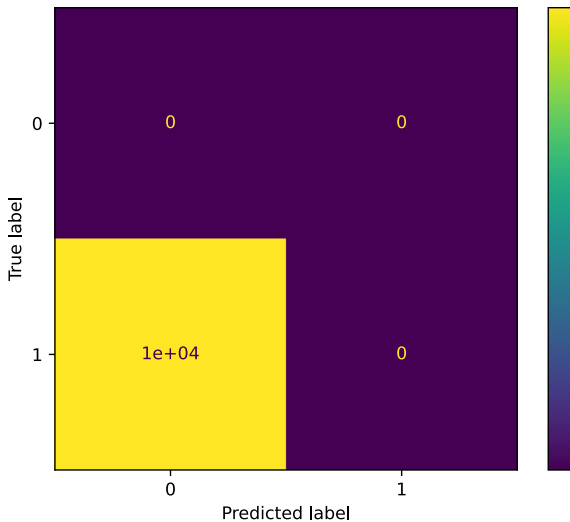
Training on All Data



Test Data Accuracy: 0.999

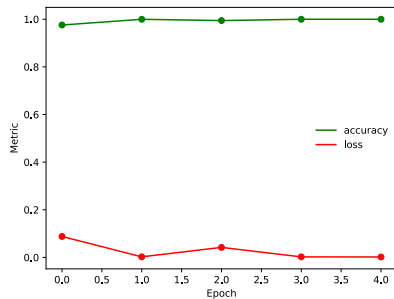
Test Data Loss: 0.007

Testing on Unseen Domain Wall Data



*Here autocorrelations may actually matter

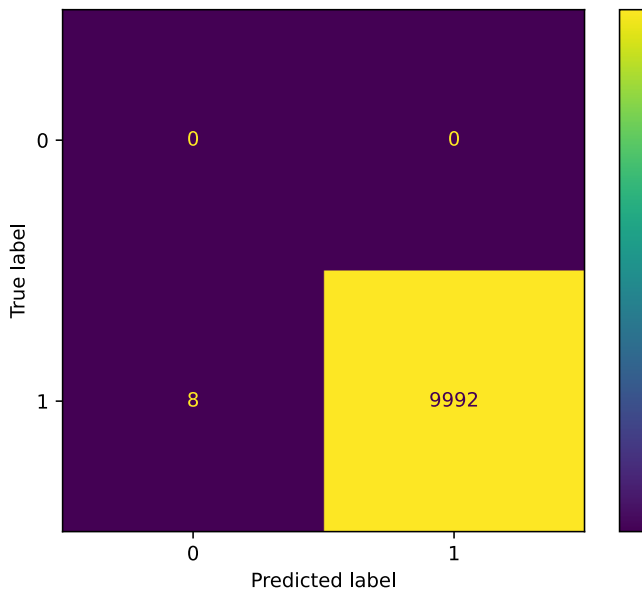
GCNN Approach



Test Data Accuracy: 0.9997

Test Data Loss: 0.002

GCNN - Testing on Unseen Domain Wall Data



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Conclusions

- Showed MLP strengths and weaknesses in classifying FM order in the Ising model
- Showed the ability for GCNN to learn magnetic domain formation
- Further analysis and care needed when producing data from Monte Carlo simulations for machine learning applications

References

M. Fey and J. E. Lenssen, Fast Graph Representation Learning with PyTorch Geometric (2024).

M. Schrauth, F. Dusel, Y. Thurn, F. Goth, D. Herdt, and J. S. E. Portela, hypertiling (2024).

I. Boettcher, A. V. Gorshkov, A. J. Kollár, J. Maciejko, S. Rayan, and R. Thomale, Physical Review B **105**, 125118 (2022).